

DNA Analysis Report

Professional-grade local genomic analysis

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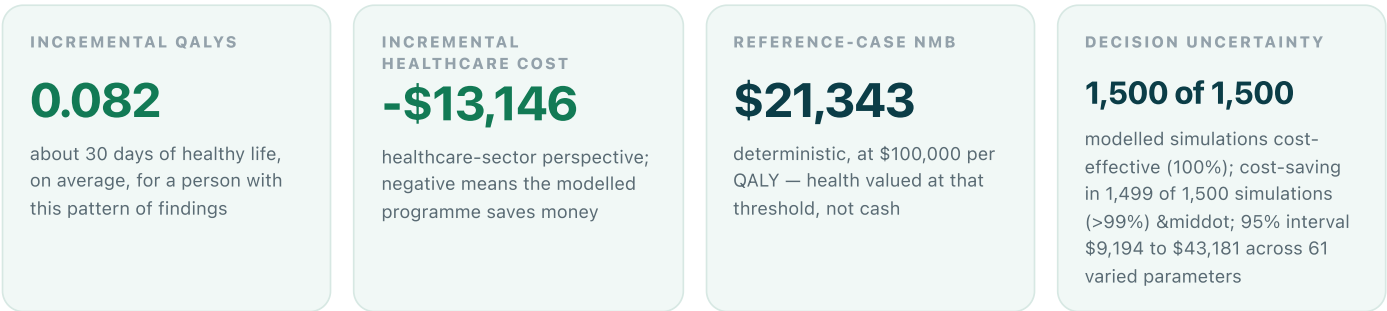
Research / Educational Use — Not a Clinical Diagnostic.

This report integrates a curated variant database, CPIC-style pharmacogenomic phenotyping, curated polygenic risk scores, compound-heterozygosity detection, carrier-status analysis, trait predictions, and optional local-AI interpretation. Findings — especially pharmacogenomic phenotypes and pathogenic variants — should be confirmed by a CLIA/CAP-certified clinical laboratory before any medical decision is made. Always consult a qualified physician, clinical pharmacist, or board-certified genetic counsellor.

Findings-first economics

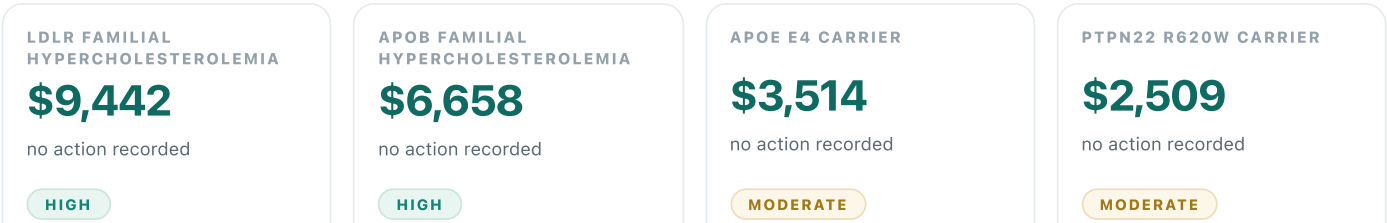
Acting on these findings is modelled to improve health and reduce healthcare spending.

This report starts with what was found in the genome. The economics explains the consequence of acting on each finding; the pooled reference case shows what remains after overlapping signals, imperfect adherence and the real cost of acting are all accounted for.



What rises to the top

Ranked by expected net monetary benefit, each valued on its own. Finding-level values are not additive; the pooled reference case is calculated separately.



Your findings at a glance

Findings that could change a decision

Every finding the model examined, the decision it bears on, the strength of the evidence, and what it is worth on its own. No dollar figure appears without the genomic result that produced it — the detail behind each figure is on pages 3 and 4.

RISK & PREVENTION 13 findings

LDLR familial hypercholesterolemia (heterozygous)	—	HIGH	\$9,442
APOB familial hypercholesterolemia (heterozygous)	—	HIGH	\$6,658
APOE ε4 carrier (elevated cardiovascular / Alzheimer's risk)	—	MODERATE	\$3,514
PTPN22 R620W carrier — autoimmune screening awareness	—	MODERATE	\$2,509
CHEK2 I157T carrier — moderate cancer risk awareness	—	MODERATE	\$2,449
Factor V Leiden / Prothrombin carrier — situational anticoagulation	—	MODERATE	\$2,273
HLA-DQ2/DQ8 carrier — celiac rule-out capability	—	MODERATE	\$2,097
predicted-pathogenic 12:111803962	Confirm the predicted variant clinically...	MODERATE	\$1,995
predicted-pathogenic 4:46312575	Confirm the predicted variant clinically...	MODERATE	\$1,893
HBB carrier (reproductive)	—	MODERATE	\$1,226
SERPINA1 carrier (reproductive)	—	MODERATE	\$1,226
G6PD deficiency — oxidative drug avoidance	—	MODERATE	\$463
Coeliac-permissive HLA haplotype (DQ2.5/DQ8) — serology decision	—	MODERATE	-\$122

LOWER-CONFIDENCE & EXPLORATORY 8 findings

OPRM1 OPRM1 A118G (mu-opioid receptor) — OPRM1 A118G...	—	LOW	\$699
LRRK2 G2019S carrier — Parkinson's early intervention	—	LOW	\$613
DRD2/ANKK1 DRD2 / ANKK1 Taq1A — DRD2/ANKK1 Taq1A — reward-pathway...	—	LOW	\$442
NAT2 — NAT2 slow acetylator — occupational cancer risk awareness	—	LOW	\$90
PON1 — PON1 Q192R — organophosphate susceptibility	—	LOW	\$90

Medication-genotype findings

Where a genotype could change a prescribing decision

These are the clearest cases of a genomic result bearing on a future medication choice. The clinical decision leads; the economics explains why preserving the genotype for that decision has value.

This run produced no medication-genotype findings with a standardised economic pathway.

Decision support, not a directive. Nothing here is an instruction to start, stop or change a medication. Relevance depends on whether the drug is actually prescribed, and the prescribing decision belongs with a qualified clinician who can see the whole clinical picture.

Risk, prevention & exploratory findings

Findings with weaker evidence, reported anyway

Lower confidence changes how a finding should be read, not whether it is shown. A negative expected NMB means the modelled cost of acting exceeds the modelled benefit — useful information, not an error. Evidence grade reflects the strength of the association, not the size of the value; provenance is on page 8.

LDLR familial hypercholesterolemia (heterozygous)

HIGH

POTENTIAL DECISION

No standardised action pathway recorded.

KEY ASSUMPTION

Baseline event probability and the effect of acting are registered assumptions varied in the sensitivity analysis; neither is measured for this individual.

EXPECTED NMB

\$9,442

medical cost averted \$9,136

health gain 0.008 QALY

cost to act \$500

APOB familial hypercholesterolemia (heterozygous)

HIGH

POTENTIAL DECISION

No standardised action pathway recorded.

KEY ASSUMPTION

Baseline event probability and the effect of acting are registered assumptions varied in the sensitivity analysis; neither is measured for this individual.

EXPECTED NMB

\$6,658

medical cost averted \$6,578

health gain 0.006 QALY

cost to act \$500

APOE ε4 carrier (elevated cardiovascular / Alzheimer's risk)

MODERATE

POTENTIAL DECISION

No standardised action pathway recorded.

KEY ASSUMPTION

Baseline event probability and the effect of acting are registered assumptions varied in the sensitivity analysis; neither is measured for this individual.

EXPECTED NMB

\$3,514

medical cost averted \$3,869

health gain 0.001 QALY

cost to act \$500

PTPN22 R620W carrier — autoimmune screening awareness

MODERATE

POTENTIAL DECISION

No standardised action pathway recorded.

KEY ASSUMPTION

Baseline event probability and the effect of acting are registered assumptions varied in the sensitivity analysis; neither is measured for this individual.

EXPECTED NMB

\$2,509

medical cost averted \$2,579

health gain 0.004 QALY

cost to act \$500

CHEK2 I157T carrier — moderate cancer risk awareness

MODERATE

POTENTIAL DECISION

No standardised action pathway recorded.

KEY ASSUMPTION

Baseline event probability and the effect of acting are registered assumptions varied in the sensitivity analysis; neither is measured for this individual.

EXPECTED NMB

\$2,449

medical cost averted \$2,579

health gain 0.004 QALY

cost to act \$500

Reported without a standardised value: 2 actionable wellness variant(s), Avoid primaquine/dapsone/rasburicase..., OPRM1-guided analgesic dosing + AUD treatment..., Dopamine-aware addiction treatment +..., Aromatic amine exposure monitoring + bladder..., Pesticide exposure avoidance + cholinesterase..., Awareness for autoimmune symptom clusters, Targeted supplementation (vitamin..., Awareness for high-risk situations (surgery..., Serum copper/ceruloplasmin monitoring + early..., Risk-aware screening schedule, Partner testing for reproductive risk..., Intensive cardiovascular risk reduction..., High-intensity statin + cascade screening, High-intensity statin therapy, Trauma-informed care + HPA-axis-targeted..., Reproductive genetics — 12 carrier... No registry-backed economic pathway exists for these

How the findings combine

Taken together, what do these findings mean?

This is the canonical reference case. Findings bearing on the same condition are combined on the risk scale, each condition's cost of illness is charged exactly once, and trial efficacy is discounted to real-world adherence before any value is credited.

INCREMENTAL
HEALTHCARE COST

-\$13,146

healthcare-sector perspective

INCREMENTAL QALYS

0.082

about 30 days of healthy life on average

REFERENCE-CASE NMB

\$21,343

at \$100,000 per QALY

CLASSIFICATION

dominant (more health, lower cost)

not defined — strategy dominates or is dominated

Condition pools — combined before value is credited

CONDITION	FINDINGS	BASELINE	RAW	POOLED	ADHERENCE	REALISED	COST AVERTED	QALYS	NMB
CAD	3	50.0%	98%	60%	50%	34%	\$7,410	0.022	\$9,339
Pathogenic	8	20.0%	290%	60%	65%	49%	\$5,111	0.043	\$8,851
Alzheimer	1	15.0%	30%	30%	35%	11%	\$1,630	0.001	\$1,516
IronOverload	3	20.0%	32%	17%	65%	12%	\$478	0.010	\$1,148
Depression	2	20.0%	18%	13%	50%	7%	\$379	0.007	\$817
Coeliac	1	20.0%	30%	30%	65%	20%	\$283	0.0001	-\$29

Two corrections, applied in order

The engine reports these as one combined reduction. They rest on different evidence and are separated here: overlap is a structural correction, adherence a behavioural one.

NAIVE AVOIDED COST

\$39,330

every finding counted independently

POOLED, AT TRIAL EFFICACY

\$25,969

34% removed — one liability, one cost of illness

REALISED BENEFIT

\$15,291

41% further removed — efficacy is not effectiveness

-\$13,361

condition pooling

-\$10,678

real-world adherence

Adherence is charged to the benefit *and* to ongoing intervention cost, but not to the one-off test cost. 0.039 QALYs are lost to imperfect adherence relative to trial efficacy.

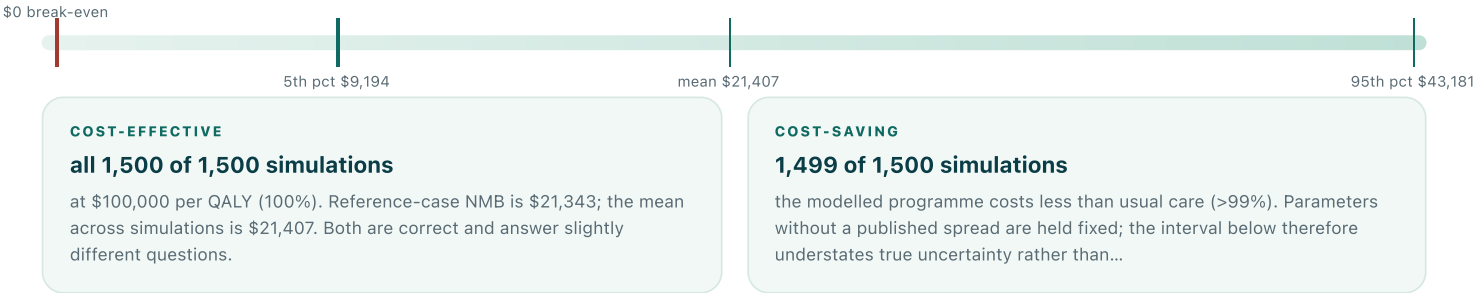
\$21,343

Uncertainty, evidence & model discipline

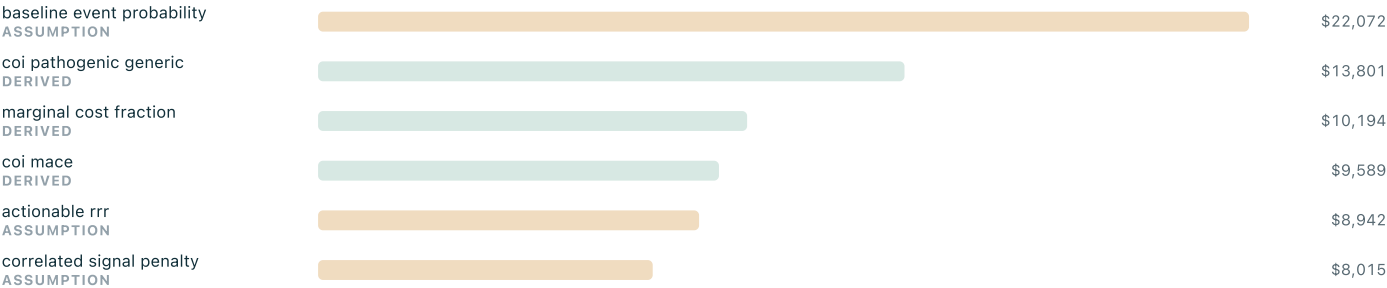
How sure is this, and what rests on judgement?

Uncertainty has two layers: the spread of modelled outcomes, and the provenance of the inputs producing that spread. A robust decision does not make its inputs precise, so both are shown.

Net monetary benefit across 1,500 simulations



What drives the answer



Swing in net monetary benefit across each parameter's plausible range. Amber bars are declared assumptions rather than sourced values; they account for 47.5% of total swing.

EVIDENCE FOUNDATION

Registry sourced	49 / 65 · 75.4%
Whole model — PMID/DOI	49.3%
Whole model — attributed	95.7%
Whole model — judgement only	4.3%

Two different denominators — 65 registered parameters and 371 known model parameters. Reported separately, because collapsing them into one figure would create an inconsistency rather than remove one.

WHAT COULD CHANGE THE ANSWER

A meaningful share of this answer rests on judgement rather than published evidence. The direction is more trustworthy than the exact numbers.

0 ERRORS report-consistency validation — rendering permitted

4 warnings · 5 information notes, listed in full in the technical appendix

The testing decision

Is whole-genome sequencing worth it?

Two different questions live on this page and they are never added together. One is a choice made *before* testing, averaged over a population. The other counts what sequencing actually found in this genome.

PROSPECTIVE — BEFORE TESTING		OBSERVED — IN THIS GENOME	
What is the expected value of choosing sequencing rather than an array?		What did sequencing actually find that an array could not?	
Chance an array misses an actionable finding	1 in 56	Sequencing-only findings present	4
Value if you are that person	\$25,189	Observed standalone contribution	\$6,340
Expected value, averaged over everyone	\$491	A count of findings this genome contains, not an expectation. For this input the population question is already settled.	
Extra cost over an array	\$200		
Net expected value	\$291		

Separation rule. The prospective figures describe the population this genome was drawn from; the observed figure describes the genome itself. They answer different decision questions, come from different inputs, and are never combined into one headline total.

Strategy frontier

STRATEGY	COST	QALYS	ICER VS PREVIOUS	STATUS
No testing	\$0	20.000	—	efficient
Genotyping chip	\$200	20.060	\$3,333/QALY	efficient
Whole-genome sequencing	\$600	20.120	\$6,667/QALY	efficient
Targeted clinical panel	\$900	20.090	—	dominated

Frontier values illustrate the testing choice for a population and are not this genome's reference case. This is an average over a population, and the distribution is extremely skewed: the great majority of people get nothing, and a small number get something that changes their care substantially. An expected value is the right...

Methods & provenance

What was assumed, where it came from, what was checked

The first seven pages tell the story; this page is their audit trail. Full parameter tables, references and the advanced analyses follow in the technical appendix.

MODEL		PROVENANCE	
Perspective	healthcare sector	Registered parameters	65
Comparator	usual care	Published	8
Outcomes	cost, QALYs, NMB	Derived	41
Time horizon	10 years	Declared assumptions	16
Discounting	3%, costs and QALYs	Registry sourced	75.4%
Willingness to pay	\$100,000/QALY	Model, resolvable	49.3%
Currency	USD, United States	Model, attributed	95.7%
Pooling	complement of products	Model, judgement only	4.3%
Adherence	benefit and ongoing cost	A parameter declared as an assumption may not cite a source — the registry refuses to load if one does.	
Uncertainty	1,500-run PSA		

Declared limitations

- Population-average parameters are applied to one genome. The model values expected consequences for a person with this pattern of findings, not a prediction for any individual.
- Estimates are illustrative and decision-analytic. This is not a formal economic evaluation or a submission to any authority.
- Where a condition-specific baseline risk is unavailable a generic registered assumption is used; those assumptions dominate the sensitivity analysis and are labelled as such.
- Costs are not inflation-normalised to a single price year.
- Findings without a registry-backed pathway are excluded from net monetary benefit rather than assigned an invented value.
- Clinically relevant findings require confirmation by an appropriate professional before anything follows from them.

Validation

- PASS** NMB identity — $\lambda \times \Delta QALY - \Delta Cost$ reconciles to the reported total

PASS Condition cost charged once — no pool exceeds its cost of illness

PASS WGS basis separation — observed and prospective never combined

PASS Report validation gate — no errors; rendering permitted
- PASS** Dominance classification — sign of cost and effect agrees with the label

PASS Pooled risk cap — combined risk reduction stays within its bound

PASS Budget-impact consistency — peak burden is the maximum net spend

Audit trail. Every figure in this report is serialised in `economics-payload.json`, the canonical object these pages render from. The parameter registry, references, cross-path reconciliation and the advanced analyses follow in the technical appendix. Build GENOMELENS-BUILD: 5b174cb 2026-09-04T17:22:06-04:00. Synthetic input — no human genome was used.